Industrial AI Predictive Maintenance Platform

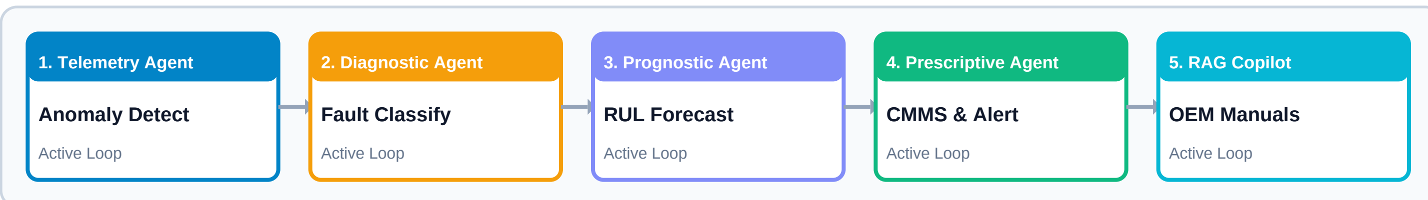
Architecture & Operational Workflow Guide (ARC.pdf) | Version 2.5 Enterprise

1. Executive Architectural Summary

This document details the complete end-to-end multi-agent architectural pipeline, mathematical formulations, and operational workflow of the **AI Predictive Maintenance Operations Platform**. Designed for high-reliability manufacturing environments (CNC machining, stamping presses, welding robotics, and conveyor lines), the system continuously ingests multi-channel IoT sensor telemetry to predict Remaining Useful Life (RUL), diagnose root-cause mechanical failure modes, and generate prescriptive ERP work orders before catastrophic downtime occurs.

2. End-to-End Multi-Agent Data Pipeline

The system operates on an autonomous 5-stage multi-agent supervisory loop:



3. Telemetry Execution: Single Step vs. Continuous Stream

A common point of operational questions is understanding why the **Synthetic Anomaly Injector Console** offers both *Single Step* and *Continuous Stream* execution modes. Here is the detailed technical breakdown:

Execution Mode	What It Does Under the Hood	When to Use It
Single Step **(1 Reading Packet)**	Sends a single instantaneous IoT telemetry packet at point *t*. The Telemetry Agent evaluates anomaly status for that exact reading snapshot.	Use when testing precise incremental metric changes or verifying instantaneous sensor threshold triggers.
Continuous Stream **(5 Progressive Steps)**	Simulates automated progressive sensor degradation over 5 sequential time steps (*t+1* to *t+5*), applying cumulative noise and thermal rise curves.	Use when observing how machine RUL drops continuously over time and testing automatic multi-step work order generation.

■ ***User Experience Enhancement Note:*** *To eliminate operational complexity, the Fault Injector console in the dashboard now features a clean* ***Simulation Execution Mode selector*** *(Single Step vs Continuous Stream) paired with a single prominent* ■ ***Execute Telemetry Injection*** *button.*

4. Core Technical Modules & Formulations

System Module	Mathematical / Algorithmic Foundation	Operational Benefit
Overall Equipment Effectiveness (OEE)	OEE = Availability x Performance x Quality *A = Operating Time / Planned Time* *P = (Ideal Cycle x Count) / Operating* *Q = Good Units / Total Units*	Quantifies exact plant-wide production efficiency grade (Target > 85%).
Signal Processing (FFT Spectrum)	X(f) = Sum x(t) * e^(-j 2 pi f t) Computes spectral power density across 0-500 Hz to detect 1X/2X misalignment frequencies.	Identifies bearing outer/inner race pass frequencies before physical failure.
Explainable AI (SHAP XAI)	Shapley values phi_i(x) scoring percentage risk contributions for each sensor metric.	Provides clear feature risk attribution (e.g. Vibration +45%, Temp +30%).
Multimodal Inspection (CV & Audio)	Optical bounding box CNN inference + Microphone decibel spectrogram peak matching.	Detects surface metal spalling, fluid leaks, and pump cavitation noise.
RAG OEM Knowledge Base	Vector Index (FAISS / Cosine Similarity) over OEM technical manuals (SKF, Siemens, Parker).	Technicians receive instant torque specs (e.g. 45 Nm) and step-by-step SOP guides.

5. Step-by-Step Operator Navigation Guide

- **Tab 1: Fleet Overview & Telemetry** — Monitor plant-wide health matrix, KPI cards, real-time Recharts waveforms, and run hardware fault simulations.
- **Tab 2: FFT Signal & SHAP XAI** — Inspect vibration frequency power spectra (0-500 Hz) and view SHAP anomaly risk attributions.
- **Tab 3: Visual & Acoustic AI** — Run computer vision optical part defect scans and microphone audio spectrogram noise checks.
- **Tab 4: CMMS Work Orders & Inventory** — Review prescriptive repair tickets, track warehouse spare parts stock, and export SAP PM / IBM Maximo JSON schemas.
- **Tab 5: RAG AI Assistant** — Ask natural language operational queries and retrieve OEM manual citations with exact page numbers and torque specs.